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MIL-S-62095C(AT)
13 January 1993
SUPERSEDING
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17 July 1987

MILITARY SPECIFICATION

SEMITRAILERS, TANK: GENERAL PURPOSE WATER, 7580 AND 15 160 LITERS (2,000 AND 4,000 GALLON), MODIFIED COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 7580 liter (L) (2,000-gallon), single axle and 15 160 L (4,000-gallon), tandem axle, tank type semitrailers for the transportation of nonpotable water. Vehicles procured under this specification are commercial items that are required to be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The semitrailer shall be one of the following types, sizes and classes, as specified (see 6.2):

Type I - Forced discharge.
Type II - Gravity discharge.

Size 2 - 7580 L (2,000 gallon), single axle.
Size 4 - 15 160 L (4,000 gallon), tandem axle.

Class 1 - Uninsulated.
Class 2 - Insulated.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage: Vehicular Ignition Lighting and Starting.

MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

* MIL-STD-209 - Slinging and Tiedown Provisions For Lifting And Tying Down Military Equipment.

MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plates Standards.

MIL-STD-1474 - Noise Limits for Army Materiel.

MIL-STD-1595 - Qualification of Aircraft, Missile and Aerospace Fusion Welders.

MS 51335 - Pintle Assembly, Towing, 18,000 Lbs. Capacity, Manual Release.

MS 75021 - Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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* DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DOT)
Federal Motor Carrier Safety Regulations (FMCSR).
Federal Motor Vehicle Safety Standards (FMVSS).
Regulations Governing Transportation of Dangerous Articles in Tank Motor Vehicles.

(Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Code of Federal Regulations, 29 CFR.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, 29 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
American National Standard Pipe Thread Series.

(Application for copies of ANSI publications should be addressed to the American National Standards institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ASME Boiler and Pressure Vessel Code (BPVC)
Section IX - Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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* AMERICAN WELDING SOCIETY (AWS)

AWS B2.1	- Standard for Welding Procedure and Performance Qualifications.
AWS D1.1	- Structural Welding Code-Steel.

(Application for copies of AWS publications should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 31025.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual.

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices.

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DOD adopted).
J350	- Spark Arrester Test Procedure for Medium Size Engines (DOD Adopted).
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation (30-1,000 MHZ).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J682	- Rear Wheel Splash and Stone Throw Protection (DOD Adopted).
J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers (DOD Adopted).
J702	- Brake and Electrical Connection Locations Truck-Tractor and Trailer.
J1292	- Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.10.4, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, sold commercially to the public, which meet or exceed the requirements of this specification. All items shall be as represented and rated in the vehicle manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The vehicle shall be a standard commercial semitrailer, of a model sold to a significant number of buyers other than the Government, in the course of normal business operations.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification markings and data plates shall be in accordance with MIL-STD-1223.

3.1.1.3 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

* 3.1.1.4 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the semitrailer manufacturer's standard matching hardware.

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* 3.1.1.5 Welders and welding. In addition to the requirements of 3.22.4, all welders employed in the fabrication of the semitrailer shall be certified, before any welding is accomplished, in accordance with MIL-STD-1595; the welding qualifications of the ASME BPVC, Section IX; or AWS B2.1 or Dl.l. The certification that the welders have passed the qualifications test shall be on file at the contractor's facility and shall be available for review by the Government.

* 3.1.1.6 Lubrication chart. An illustrated or schematic diagram type lubrication chart shall be provided on the semitrailer. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the semitrailer in a readily visible location. The chart shall be inscribed on a plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514.

* 3.1.1.7 Lifting and tiedown attachments. When specified (see 6.2), the vehicle shall have lifting and tiedown attachments. Lifting and tiedown attachments shall be in accordance with class 1 and 2 or in accordance with class 3 of MIL-STD-209. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209.

* 3.1.1.7.1 Tiedown identification. Tiedown attachments shall be identified by stenciling or other suitable marking. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the vehicle on the carrier.

* 3.1.1.7.2 Transportation plate. Lifting and tiedown provisions shall include a transportation plate conforming to 3.1.1.8. The transportation plate shall have a diagram showing the attachments and proposed lifting slings, the capacity of each attachment, and the required length and size of each sling cable. The transportation plate shall have a silhouette of the vehicle showing the center of gravity.

* 3.1.1.8 Information plates. The transportation plate specified in 3.1.1.7.2, the sign specified in 3.2.8, the caution plate in 3.16.2.1 and the instruction plate in 3.16.5 shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

* 3.1.1.9 Pintle assembly. When specified (see 6.2), a trailer towing package shall be furnished. The trailer towing package shall consist of a pintle, safety chain attachment devices, an electrical receptacle and associated reinforcements and wiring and shall be installed on the rear of the vehicle. The pintle shall conform to MS 51335-2. The pintle shall be installed in the chassis frame with reinforcements to transfer a vertical tongue load of not less than 815 kilograms (kg) (1,800 pounds) and a horizontal drawbar load of not less than 8150 kg (18,000 pounds) directly

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to the chassis rails. The rearmost portion of the pintle shall be forward, but (four inches) forward, of the rearmost part no more than 100 millimeters (mm) of the vehicle. The height of the pintle assembly measured from the ground to the lower inner jaw of the pintle shall be 787 mm (31 inches) ±25 mm (±l inch) with the semitrailer loaded to its rated payload capacity.

* 3.1.1.9.1 Safety chain attachment devices. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength of not less than 8150 kg (18,000 pounds). The attachment devices shall be capable of accommodating a standard grab hook [56 mm (2-3/16 inches) wide, 18.7 mm (47/64 inch) thick and 12.3 mm (31/64-inch) throat width)] for a 9.5 mm (3/8-inch) chain.

* 3.1.1.9.2 Electrical receptacle. An electrical receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein shall be installed near the pintle. The receptacle shall be provided with a spring loaded cover.

* 3.1.1.10 24-volt lighting.

* 3.1.1.10.1 Independent 24-volt dc system. When specified (see 6.2), a 24-volt direct current (dc) lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.13. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.13.2, except the 24-volt dc stop and turn lamps shall be combined.

* 3.1.1.10.2 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the 12-volt dc System (see 3.13.2) from a towing vehicle equipped with a nominal 24-volt dc electrical system shall also be provided.

* 3.1.1.10.3 Receptacle, 24-volt dc. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B.	Connect to left-hand turn signal and stop lamp (yellow).
Contact D.	Connect to ground (white).
Contact E.	Connect to clearance, side marker, identification, and tail lamps (black and brown).
Contact J.	Connect to right-hand turn signal and stop lamp (green).
Contact L.	Connect to ground (white).

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The rest of the contacts shall not be connected. Circuits B and J on tactical trucks are combination stop and turn indicator circuits. On the interconnected 24-volt system, the normal 12-volt de turn signal lights will function both as turn signals and stop lights, and the normal 12-volt de stop lights and any antilock brake components will not be operational when the semitrailer is connected to a towing vehicle with a 24-volt de power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt dc, 12-contact receptacle.

* 3.1.1.10.4 Resistors. When an interconnected 24-volt de system is specified, the 12-contact receptacle shall be provided with resistance in each circuit to reduce the voltage of the tactical (military design) towing vehicle from a nominal 24 volts de to 12 volts dc. Each circuit resistor shall be selected to reduce the 28-volt dc regulated voltage of the tactical truck to be compatible with the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any damage to resistors, wiring, or adjacent components.

3.2 General design.

* 3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

* 3.2.2 OSHA and FMCSR regulations. The vehicle, with only furnished attachments, accessories and equipment, shall enable compliance by any user with all OSHA and FMCSR user regulations that are vehicle or vehicle equipment dependent. Regulations shall be those that are applicable to a commercial user of such a similar type, class and size of semitrailer for the same general use. See 6.1.

* 3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When ratings are not published, verification of ratings shall be available to the engineering office of the procuring activity.

* 3.2.4 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

* 3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

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* 3.2.6 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.7 Lubrication. Lubrication means shall be provided for all parts of the semitrailer normally requiring lubrication. Where the use of high lubricating pressure will damage grease seals or other parts, fittings with pressure release shall be used.

* 3.2.8 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather and all rotating or reciprocating parts which are of such a nature or so located as to become a hazard to operating personnel shall be insulated, fully enclosed, or properly guarded. A sign shall be attached in a suitable location near the discharge line(s), with a minimum of 50 mm (2-inch) high letters, stating: "WATER IS NOT FIT FOR DRINKING." The sign shall conform to 3.1.1.8.

* 3.2.9 Radio interference suppression. The engine on type I forced discharge semitrailers shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 3.3 Performance. The semitrailer shall evidence no part failure, deformation, permanent set or interference between parts when towed, both empty and when loaded as specified in 3.4:

(a) At speeds as great as 32 kilometers per hour (km/h) (20 miles per hour (mph)) over unimproved roads and reasonably hard uneven terrain;
(b) At speeds as great as 97 km/h (60 mph) over improved roads; and
(c) At speeds as great as 16 km/h (10 mph) on 10 percent side slopes on unimproved roads and reasonably hard uneven terrain.

3.3.1 Turning ability. The semitrailer shall be capable of assuming a 90-degree angle to a truck tractor without cramping or damage to the semitrailer or the truck tractor.

3.3.2 Tracking ability. The semitrailer shall conform to the tracking ability requirements of Federal Motor Carrier Safety Regulation 393.70(a) when operated under the conditions specified in 3.3.

* 3.3.3 Service brakes. The service brakes shall control, decelerate and stop the semitrailer-tractor combination, with the semitrailer loaded with specified payload, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52. Requirements shall be met with a truck tractor loaded to its full rated gross vehicle weight (GVW) and gross combination weight (GCW).

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3.4 Payload capacity. The rated payload capacity shall be not less than the weight computed from the rated liquid volume, using 1.0 kilogram per liter (KG/L) (8.3 pounds per gallon). Load bearing members shall be rated for the loads imposed with the tank in a horizontal position, and the liquid payload level in the tank.

* 3.4.1 Total axle load. The gross weight (curb weight plus payload) on the axle(s) shall not exceed 9070 kg (20,000 pounds) for single axle semitrailers, and 15 400 kg (34,000 pounds) for tandem axle semitrailers.

3.5 Dimensions and clearances. The semitrailer, uncoupled from the truck tractor, resting on its landing legs on level ground, without payload, and with the tank body within 1 percent of level, shall conform to the dimensions specified in table I.

TABLE I. Dimensions and clearances.

Characteristic	mm (Dimension or clearance)
Overall length (maximum)	9140 (30 feet) for size 2 9750 (32 feet) for size 4
Overall width (maximum)	2440 (96 inches)
Ground clearance (minimum)	200 (11 inches)
Upper fifth wheel height, from the ground to the underside of the upper fifth wheel plate, +25 mm (+I inch)	1220 (48 inches)
Distance from vertical centerliner of kingpin to front end of semitrailer (not less than)	610 (24 inches) for size 2 910 (36 inches) for size 4
Landing wheel clearance - radius from vertical centerline of kingpin to any point 150 mm (6 inches) below upper fifth wheel plate (minimum)	2130 (84 inches)
Swing radius from vertical centerline of kingpin to most distant point on semitrailer nose (maximum)	1330 (52.5 inches) for size 2 1420 (56 inches) for size 4

* 3.6. Frame. The semitrailer shall be of the frame or frameless type.

* 3.6.1 Stress analysis. The contractor shall perform a stress analysis. The analysis shall verify compliance with the requirements specified and shall include at least the following calculations and diagrams:

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(a) The sprung gross weight distribution diagram;
(b) The vertical shear diagram for the length of the vehicle;
(c) The bending moment diagram for the length of the vehicle;
(d) The loads, sections, and stresses in members that carry maximum shear and moment, to include the analysis of the stresses in attaching weldments or other fastenings;
(e) The loads, sections, shear, and bending moment diagrams at the kingpin and rear-underconstruction subframe members and for the drop section, when drop frame semitrailers are provided;
(f) The load, section, shear, and bending moment diagram for the most critical crossmember, to include an analysis of stresses in attaching weldments or other fastenings;
(g) The loads, sections, and stresses in members that carry maximum compression;
(h) A listing of all assumptions and references used to provide required data.

* 3.7 Axles. Size 2 semitrailer shall be provided with a single axle. Size 4 semitrailer shall be provided with tandem axles. The axle rating shall be at least equal to the load imposed on the axle(s), measured at the ground, with the semitrailer loaded as specified in 3.4. The wheel bearings and axle spindles shall be oil lubricated. The hubs shall be fitted with seals. The hub caps shall have a window and shall incorporate provisions for venting or an equivalent method of withstanding internal pressure buildup without leakage.

3.8 Suspension system. The semitrailer shall be equipped with a suspension system, with components having a rated capacity at least equal to the load imposed upon each member, measured at the ground, with the semitrailer loaded as specified in 3.4. When the suspension system is rated at the spring pads, unsprung weight shall be deducted.

3.8.1 Full air suspension. When specified (see 6.2), the semitrailer shall be equipped with a full air suspension system, with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the semitrailer loaded as specified in 3.4. Hydraulic double-acting shock absorbers shall be provided on the axle assembly. The suspension shall be provided with a mechanism at each wheel to assure lifting of the wheel and axle when jacking the semitrailer from the applicable jacking location. The suspension system shall incorporate leveling valves with time delay to avoid constant air consumption. Shock absorbers and springs shall be fabricated of materials capable of operating in ambient -46 degrees Celsius (°C) (-50 degrees Fahrenheit (°F)) to +52°C (+125°F). The suspension system shall have none of the sprung weight carried on anything other than the air springs. No other spring medium, such as leaf, coil, or rubber springs, shall be used in conjunction with the air springs.

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* 3.9 Wheels, rims and tires. The semitrailer shall be equipped with dual disc type wheels on each axle. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Multi-piece rims shall not be furnished. Rim sizes shall be the same for all wheels on the semitrailer.

* 3.9.1 Tires. Steel belted radial, tubeless tires with highway tread shall be furnished. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the semitrailer loaded as specified in 3.4. Tire size and load range (ply rating) shall be the same for all tires on the semitrailer. Tires shall conform to Tire and Rim Association or to European Tyre and Rim Technical Organisation recommendations.

* 3.9.2 Carrier for spare tire assembly. A carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the curbside, forward of the rear axle(s). The carrier design shall enable safe removal or mounting of a spare wheel assembly using only tools furnished with the semitrailer. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the semitrailer and carrier without personnel positioning themselves or any part of their body under the spare tire assembly. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion resistant material.

3.9.3 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be mounted and secured on the carrier. When a spare tire assembly is specified (see 3.9.4), a spare wheel or rim shall be furnished.

3.9.4 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the vehicle.

* 3.9.5 Tools. When specified (see 6.2), the semitrailer shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel, with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

* 3.10 Brakes. Full air, internal expanding type brakes shall be furnished. The brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, 393.45 through

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393.47, and 393.52. Brake linings shall be of nonasbestos material. The brake system shall include:

(a) Piping
(b) Hose connections
(c) Gladhands
(d) Retracting rubber diaphragm or spring loaded dust covers or dummy gladhands equipped with security chains or cables
(e) Provisions for stowing dummy gladhands
(f) Relay emergency valve
(g) Air reservoirs with a remote control drain located above the bottom of the frame rails
(h) Automatic slack adjusters
(i) Brake chambers and
(j) All other parts and connections required for a complete air brake system.

Gladhands shall conform to SAE J318. The location of the gladhands shall be in accordance with SAE J702. All components of the brake system shall be installed in a manner to provide road clearance for travel over uneven or rough terrain and protection from objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of the wheel rims.

* 3.10.1 Parking brakes. The semitrailer shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall hold the semitrailer, with rated payload, on a 10 percent grade despite the depletion of the compressed air supply. The parking brakes shall be automatically applied upon disconnection of the emergency air line and under emergency conditions. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

* 3.11 Upper fifth wheel. The upper fifth wheel plate shall completely cover a 910 mm (36 inch) diameter lower fifth wheel when the lower fifth wheel is coupled to the kingpin of the upper fifth wheel and the semitrailer is turned at all angles. The upper fifth wheel shall conform to Federal Motor Carrier Safety Regulation 399.70(b).

* 3.11.1 Upper fifth wheel plate. The upper fifth wheel plate shall be not less than 6.4 mm (0.25 inch) thick. The plate shall be reinforced on the upper surface through the use of commercial steel angles, channels and ribs. The plate shall include an access hole to permit servicing the brake and electrical connections when these components are not serviceable from the top.

* 3.11.2 Pickup plate. A turned up lip pickup plate, compatible with the size (payload) of the semitrailer, shall be provided as an extension of the upper fifth wheel plate. The pickup plate shall

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have a turned up lip not less than 50 mm (2 inches) high. The pickup plate shall extend sufficiently forward to protect the semitrailer from damage during coupling and uncoupling operations to and from a truck tractor. The pickup plate shall be integral with the upper fifth wheel plate.

* 3.11.3 Kingpin. A 50 mm (2-inch) diameter kingpin conforming to SAE J700 shall be furnished. The kingpin shall be fabricated of heat treated alloy steel.

* 3.12 Landing gear. The semitrailer shall have two vertical lift, telescopic, nonrotating, landing legs with two-speed gears, and a hand crank on the curbside. Landing legs shall be provided with self-leveling skid pads of not less than 250 mm by 250 mm (10 inch by 10 inch) square bearing area. A holder shall be provided for the hand crank when not in use. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to the portion of the gross weight sustained, and the additional forces resulting from impact during coupling and uncoupling operations. When placed in the travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to prevent the entrance of foreign matter which would impair its functioning or mechanical efficiency. The semitrailer shall not nose over when resting on the landing gear.

* 3.12.1 Landing gear range of adjustment. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel plate at the kingpin from 1190 mm (47 inches) to not less than 1320 mm (52 inches) from the ground for the size 2 semitrailer, and from 1190 mm (47 inches) to not less than 1370 mm (54 inches) from the ground for the size 4 semitrailer. The minimum ground clearance with the landing gear retracted shall be 280 mm (11 inches).

* 3.12.2 Air/hydraulic powered landing gear. When specified (see 6.2), in addition to the requirements of 3.12 and 3.12.1 (less hand crank, etc.), the landing gear shall be air/hydraulic powered. The landing gear system shall include a control box equipped with a 12-volt dc motor driven pump, master control valve, leveling valve, booster, main oil tank, air side oil tank, key switch and mounting hardware.

* 3.13 Lighting. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of rub rails (unless recessed and fully protected) or on the semitrailer bumper. All electrical wiring shall conform to SAE J1292. Clearance and identification lights shall be constructed for easy removal and replacement of lamps and lenses without the use of any tools.

3.13.1 Twelve-volt direct current (dc) system. A 12-volt dc lighting system conforming to 3.13 and 3.13.2 shall be provided. The 12-volt dc lighting system shall conform to Federal Motor

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Carrier Safety Regulations 393.9, 393.11, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33.

3.13.2 Receptacle for 12-volt dc system. The front of the semitrailer shall be equipped with a 7-contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. the receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.14 Rear end protection. Semitrailer rear end protection shall be provided in accordance with Federal Motor Carrier Safety Regulation 393.86.

* 3.15 Tank body and equipment. The semitrailer shall be designed and constructed to meet the requirements of DOT Safety Regulation 178.340, MC 306 or, at the manufacturer's option, DOT Safety Regulation 178.345, DOT 406 for material, structural integrity, joints, supports and anchoring, circumferential reinforcement, and minimum thickness, except in no case shall the minimum thickness of any part of the tank be less than 12 gage (2.657 mm) (0.1046 inch). Minimum thickness requirements shall be based on a product density of 1.0 kilogram per liter (kg/L) (8.3 pounds per gallon).

* 3.15.1 Material. The tank shell, baffles, bulkheads, pads, manholes, and outlets shall be of type 304 stainless steel. Thickness of the shell, head, and baffle sheets shall be not less than 12 gage (2.657 mm) (0.1046 inch) stainless steel. Tank combinations of mild steel, high strength low alloy steel, and austenitic stainless steel, as permitted by MC 306, shall not be furnished. All hose couplings, dust caps, dust plugs, dust cap and dust plug security chains or cables, valves, piping, pipe fittings, tubing and strainers shall be of stainless steel. Vents shall be of aluminum with a stainless steel weld ring with the aluminum isolated from contact with steel, or shall be of stainless steel. Faucets and nozzles shall be of stainless steel or brass. Teflon packing and tape shall not be used on or adjacent to parts that have an aluminum content.

3.15.2 Capacity. The liquid capacity of the tank shall be not less than 7580 L (2,000 gallons) for size 2 semitrailers and 15 160 L (4,000 gallons) for size 4 semitrailers, both plus at least 3 percent expansion space by volume. The tank shall have a single compartment.

3.15.3 Tank configuration. The tank shall have an elliptical, circular, or modified rectangular section. The tank body may be provided with a slope to ensure the complete removal of water. When a slope is provided, it shall not exceed one percent of the tank body length.

* 3.15.4 Baffles. The baffle requirements of MC 306 or DOT 406 for circumferential reinforcements and surge protection shall be met. Baffles shall be flanged and dished. A half round or full round opening of not less than a 75 mm (3-inch) radius shall be provided at the

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bottom of the baffles. Openings of equivalent size shall be provided at the top of the baffles. Each baffle shall also be provided with not less than a 460 mm (18-inch) diameter manway opening. The edges of the manway opening shall have a flange of approximately 25 mm (1 inch).

* 3.15.5 Manhole. The tank shall be equipped with a round or oval manhole opening not less than 480 mm (19 inches) in major diameter. The manhole sides and lip shall be constructed of stainless steel having a carbon content that in not more than that in type 304 stainless steel. A lightweight, liquid-tight, easy opening, self-latching manhole cover, or a bolted cover with a liquid-tight, easy opening, self-latching fill cover not less than 250 mm (10 inches) in diameter, shall be supplied. The fill opening shall be provided with a removable sleeve screen of sufficient length and of a design to permit the flow of water at a rate of 950 liters per minute (L/min) (250 gallons per minute (gpm)). Hinged covers shall be hinged at the forward edge. All metal parts of the covers exposed to contact with water in the tank shall be fabricated of type 304 stainless steel.

* 3.15.6 Vents. Air vents shall be provided to allow for expansion of the water in the tank and discharging the tank at the maximum rate of 950 L/min (250 gpm)) with the manhole and fill cover sealed. The vents shall be adequate to limit the tank pressure and vacuum under these conditions to those permitted by MC 306 or DoT 406. The vents shall remain sealed when not fulfilling their purpose of equalizing internal and external pressures.

3.15.7 Catwalk and ladder. A catwalk not less than 1830 mm (72 inches) long and not less than 610 mm (24 inches) wide shall be provided over the top of the tank or on the curbside of the tank. The catwalk shall be centrally located at the manhole. The catwalk shall be of stainless steel grip-strut grating. An access ladder or steps shall be provided for access to the catwalk. The ladder or steps shall be of stainless steel.

* 3.15.8 Fenders and mud flaps. The semitrailer shall be provided with fenders encircling at least the top half of the wheels. Rubber mud flaps shall be installed to the rear of the wheels. The mud flap installation shall conform to SAE J682. A metal strip not less than 3.2 mm (0.125 inch) thick and not less than 25 mm (one inch) wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.25 inch) wide by 2.4 mm (0.094 inch) thick shall be furnished at each bolt.

3.15.9 Underslung cabinets. Underslung cabinets, constructed of steel not less than 16 gage (1.519 mm) (0.0598 inch) in thickness, shall be provided on both sides of the semitrailer, extending from the rear wheelhousing forward to the gooseneck area. The cabinets shall be of the full flanged flush type and of a size to house the equipment specified. The cabinets shall have

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provisions for flushing, draining, and ventilating through fixed louvers. The cabinets shall be stormproof. Each cabinet shall be provided with a light. Racks or shelves for stowing the discharge hose shall be provided.

3.15.9.1 Cabinet doors. The cabinet doors shall be side-hung and equipped with hinges. The hinges shall have stainless steel hinge pine and stainless steel, bronze, nylon, or nylon insert sockets. The cabinet doors shall be provided with rubber bumpers and door retainers for keeping the doors in full open position when required. The doors shall have latches and locking handles, keyed alike.

3.15.10 Hose tubes. Stainless steel hose tubes of the dimensions required to stow the intake hose specified in 3.15.11 shall be provided in a convenient location on the tank body. The hose tubes shall have drainage provisions and a removable cover at the front to facilitate cleaning. Locking devices shall be provided on each tube cover, keyed the same as the cabinet doors.

* 3.15.11 Intake (suction) hose. Four lengths of intake suction hose, each not less than 4570 mm (15 feet) in length and 75 mm (3 inches) in inside diameter (id) shall be furnished. Hoses shall withstand handling and usage in ambient air temperatures between +570 Celsius (°C) (+135 degrees Fahrenheit (°F)) and -40°C (-40°F). One male and one female coupling shall be furnished on each length of hose and shall be of the quick disconnect cam-locking type. The suction hose shall be stowed in the hose tubes.

* 3.15.12 Discharge hose. Three lengths of single jacket discharge hose, each 7600 mm (25 feet) in length and 64 mm (2.5 inches) id, shall be furnished. Hoses shall withstand handling and usage in ambient air temperatures between +57°C (+135°F) and -40°C (-40°F). One male and one female coupling shall be furnished on each length of hose and shall be of the quick disconnect cam-locking type. Discharge hose shall be stowed on racks or shelves in the side cabinet (see 3.15.9).

3.15.13 Piping. All piping shall be of not less than 89 mm (3.5 inch) nominal diameter. Piping shall have long sweep bends and shall be installed to resist vibration without failure. All threads shall be of the American National standard pipe thread series.

3.15.14 Gage. A float type water gage shall be provided at the rear, near the curbside of the semitrailer. The gage shall be calibrated in both increments of 500 L and in increments of 100 gallons and shall indicate the actual volume of water in the tank within 378 L (100 gallons). The sight cylinder shall be of the transparent Plexiglas type material and shall be protected against breakage. Valves shall be provided to seal off the gage in the event of gage failure or malfunction.

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3.16 Type I (forced discharge). Type I semitrailers shall be equipped with a forced discharge system and shall conform to the requirements specified in 3.16.1 through 3.16.5.

3.16.1 Pump. An engine driven, self-priming centrifugal pump shall be installed in the right (curbside) cabinet. The pump shall have 75 mm (3-inch) auction and discharge ports. The pump shall be constructed of either cast iron or welded steel. The pump shall be equipped with marked drain plug(s) located to assure complete drainage.

3.16.1.1 Suction. The engine driven pump shall be capable of filling the tank from a nonpressurized outside source at a rate of at least 756 L/min (200 gpm). Pump performance requirements shall be met with the pump drawing water from a source 4570 mm (15 feet) below the suction port of the pump, through 14 meters (m) (45 feet) of 75 mm (3-inch) id suction hose equipped with a foot strainer and with the suction hose coupled to the pump by a 45-degree elbow. Pump performance shall be met under these suction conditions while loading the tank through the manhole or fill cover with 15 m (50 feet) of 64 mm (2.5 inch) id discharge hose.

3.16.1.2 Discharge. The pump shall be capable of discharging the contents of the tank at a rate of not less than 756 L/min (200 gpm) against a static head of 7600 mm (25 feet) plus the static head created by discharge through the manifold piping and valve(s) and 23 m (75 feet) of 64 mm (2.5 inch) discharge hose.

* 3.16.2 Engine. The pump shall be driven by a diesel engine. The engine shall be air cooled, with the power, torque and speed necessary to satisfactorily operate the pump. The engine shall be furnished complete with all accessories necessary for operation, including:

(a) Air cleaner
(b) Starting system
(c) Alternator or generator
(d) Battery
(a) Fuel tank of sufficient capacity for eight hours of normal operation
(f) Exhaust system
(g) Engine mounted instrument panel with throttle control
(h) Start/stop switch and
(i) Battery charging indicator.

The battery shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131. The engine shall be provided with an ether aid cold start system, glow plug or grid heater.

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* 3.16.2.1 Noise level. If the noise level at the operator's position, when measured in accordance with MIL-STD-1474, exceeds 84 decibels (dB(A)), a warning plate stating "CAUTION: HEARING PROTECTION REQUIRED" shall be permanently affixed to the semitrailer in a conspicuous location at the operator's position. The noise level shall be determined with the vehicle stationary and the pumping system operating at a maximum engine governed speed and maximum load. The plate shall conform to 3.1.1.8.

3.16.2.2 Exhaust system. The exhaust system shall be furnished with a spark arresting muffler. The muffler shall have a heat shield. The tailpipe shall be pointed toward the rear end and to the ground. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

* 3.16.2.3 Lights. In addition to the other lighting specified herein, lights with guards and toggle switches suitable for illuminating the engine control panel shall be furnished.

* 3.16.2.4 Engine control panel. The engine throttle and a key-operated starting switch shall be installed in a weather- and waterproof cabinet on a control panel, each identified as to their function. The cabinet shall be fitted with a side hinged cover with a spring type latch and shall provide for locking with a padlock.

3.16.3 Water tank outlet valve and controls. A tank outlet valve of the flanged type not less than 75 mm (3 inches) in nominal diameter shall be provided. The outlet valve, when closed, shall effectively seal the tank and shall be of a type permitting complete removal of the valve body. Controls with slotted stops to hold the outlet valve at various openings shall be furnished and located in the manifold cabinet. The outlet valve shall have a screen of small mesh that will not unduly restrict the flow of water through the discharge piping. The screen shall be readily removable for cleaning and shall be located on the inside of the tank over the valve mechanism.

3.16.4 Manifold. A manifold shall be provided having 75 mm (3-inch) gate valves and with cam-locking connections designed to fit the hoses specified herein, arranged to provide the following:

(a) Gravity discharge of tank contents, bypassing the pump.
(b) Pressure discharge of tank contents by use of the pump unit.
(c) Filling of the tank through the filler opening, by utilizing the pump unit and the hoses specified herein, from a non-pressurized source.

* 3.16.5 Instruction plate. A data plate, showing a schematic of all valves, components and accessories with operating instructions, shall be fastened in a visible location inside the curbside cabinet door. The plate shall conform to 3.1.1.8.

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3.17 Type II (gravity discharge). Type II semitrailers shall be designed for gravity discharge and discharge by suction at a rate of 950 L/min (250 gpm) from an outside source (fire truck pump). The semitrailer shall be equipped with tank outlet valve(s) with valve control(s) and valve screen(s) as specified in 3.16.3. Tank outlet valve control(s) shall be located in the right (curbside) cabinet. Gate valves as specified in 3.16.4 shall be provided in the right side cabinet and at the rear of the semitrailer in convenient, protected locations.

3.18 Insulation, class 2 semitrailers. The tank and all piping on class 2 semitrailers shall be completely covered with a minimum of 50 mm (2 inches) of fiberglass insulation material. External rings shall be welded to the exterior of the inner tank to provide insulation space between the inner tank and the outer insulation jacket. The insulation jacket shall be of type 304 stainless steel and shall completely enclose the tank, including the front, rear, top, sides and bottom. The insulation jacket shall be waterproof. A watertight double cover with insulation shall be provided over the manhole.

3.19 Stowage of on-vehicle equipment. Provisions shall be made for securing all equipment on the semitrailer.

3.20 Tools. Such tools as are normally required for operation of the equipment shall be furnished and stowed in the left (streetside) cabinet.

3.21 Servicing and adjusting. Prior to acceptance of each vehicle by the Government, the contractor shall service and adjust each vehicle and its mounted equipment for operational use, including at least the following:

(a) Adjustment of engine;
(b) Adjustment of brakes;
(c) A check for proper functioning of all lights;
(d) Inflation of all tires; and
(e) Complete lubrication with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.22 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Bolted, riveted and welded construction used shall be in accordance with the highest standards of the industry.

* 3.22.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to ensure uniformity of size and shape.

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* 3.22.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

* 3.22.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.22.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. See 3.1.1.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the

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contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of the product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine the curb weight and the distribution of the curb weight on the kingpin and the rear axle(s). The imposed loading on the kingpin and rear axle(s) shall be computed by the contractor and verified by the Government, using the curb weight and the payload (see 3.4) distributed in the tank. The total calculated imposed load on the rear axles shall be compared to the rated capacities of the suspension, axles, and tires to determine if these components are of adequate capacity to meet contractual requirements.

4.3.2 Leakage test. Each completely assembled tank body, including all piping, shall be tested for leaks at their design pressure in accordance with DoT Safety Regulation 178.341, MC 306. The tank of the first production vehicle shall be tested for leaks both before and after the road test.

* 4.3.3 Road test. The first vehicle for each contract line item number (CLIN) under contract shall be road tested with and without payload. The road test shall be for the distance required in table II, corresponding to the number of vehicles being procured under the specific contract line item number (CLIN). The test shall be conducted by the contractor under the direction of Government representatives. The test shall be conducted over test course approved by the procuring activity and meeting the requirements of table III. The vehicle shall be inspected periodically during and at the completion of the test by the contractor.

* TABLE II. Road test distance.

Number of vehicle being procured	Road test distance km (miles)
1 - 10	160 (100)
11 - 20	320 (200)
21 - 30	480 (300)
31 - 40	640 (400)
41 - 50	800 (500)
51 - 100	1600 (1000)
101 - up	8050 (5000)

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TABLE III. Test course requirements.

Type of road	Road test distance
Dirt	10 percent
Gravel	30 percent
Improved pavement	60 percent

* 4.3.3.1 Test course. The total test mileage shall be on roads and highways within the general geographic limits of the place of manufacture and adjoining states. Any special highway permits required shall be obtained by the contractor. The course shall be arranged to return the vehicle to the manufacturer's plant at the completion of each 1600 km (1,000 miles), if applicable, or at the completion of the test otherwise, for inspection and chassis servicing.

* 4.3.3.2 Test loads. The semitrailer shall be tested for 90 percent of the mileage with the semitrailer loaded as specified in 3.4 and for 10 percent of the mileage with the semitrailer empty. Average speeds during all road tests shall be as close as possible to those specified in 3.3.

* 4.3.3.3 Road test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new semitrailer or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test semitrailer shall be for damage or deficiencies, including but not limited to the following;

(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance

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(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

* 4.3.4 Brake tests. Brake tests to determine conformance to requirements shall be conducted during the road test. The brake tests shall be conducted with the fully loaded semitrailer coupled to a truck tractor. The truck tractor shall have a gross vehicle weight rating (GVWR) and gross combination weight rating (GCWR) not exceeding that specified in table IV. The truck tractor shall be without excessive brake capacity. GVWR and GCWR shall be the maximum ratings on the tractor only and not necessarily the actual maximum GVW and GCW of the combination of test vehicles. Failure of the vehicle combination to stop within the required 12 200 mm (40 feet) from a speed of 32 km/h (20 mph) shall be cause for rejection.

TABLE IV. Maximum tractor characteristics.

Size of semitrailer	Max. tractor GVWR kg (pounds)		Max. tractor GCWR kg (pounds)	
1	12 700	(28,000)	22 700	(50,000)
2	14 500	(32,000)	29 500	(65,000)

* 4.3.5 Turning ability test. A truck-tractor shall be coupled to the semitrailer and driven through turns up to 90 degrees to the towed vehicle, right and left, to determine conformance to 3.3.1. At the conclusion of five turns in each direction, the tractor and the semitrailer shall be examined for evidence of binding, misalinement or other damage.

* 4.3.6 Parking brake test. To test parking brakes, the semitrailer, with rated payload, shall be placed on a grade, first headed up the grade and then headed down the grade, and tested to determine conformance to 3.10.1. These tests shall be conducted with the semitrailer disconnected from the towing vehicle and resting on the landing legs.

* 4.3.7 Landing leg test. With the semitrailer disconnected from the towing vehicle and loaded as specified in 3.4, the landing legs shall be raised and lowered to vary the height of the upper fifth wheel over the landing gear's full range, to determine conformance to 3.12 and 3.12.1. The loaded semitrailer shall be coupled and uncoupled no less than 5 times and the landing legs shall be examined for binding or misalinement.

* 4.3.8 Electrical test. An electrical test shall be conducted to determine that all vehicle lights are functioning. In addition, when an inner connected 24-volt system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a

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time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 25°C (77°F). After the test, the resistor assembly, including resistors, wiring and adjacent components shall be examined for evidence of overheating, deterioration or damage.

* 4.3.9 Lifting and tiedown test. Lifting and tiedown provisions, markings, and the connecting structural members of the vehicle shall be inspected for conformance with MIL-STD-209. All lifting and tiedown provisions and the connecting structural members shall meet the criteria of MIL-STD-209. Test results that show the provisions do not meet the criteria of MIL-STD-209 shall constitute failure of this test.

4.3.10 Type I pump test. Type I semitrailer pumping equipment shall be tested by loading and unloading the tank under the conditions and at the flow rate specified in 3.16.1.1 and 3.16.1.2.

4.3.11 Type II loading and unloading test. Type II semitrailer shall be loaded and unloaded at the rate of up to 948 L/min (250 gpm). The fire hose connections, valves, and piping shall be tested by a fire truck or suitable pump drawing from the semitrailer while pumping at 948 L/min (250 gpm). Failure of the suction connections to deliver a water flow of 948 L/min (250 gpm), failure to deliver the complete contents of the tank, or damaged piping or a damaged tank due to vacuum pressure shall be cause for rejection.

* 4.3.12 Certification. When an air suspension system is furnished, the contractor shall certify capability of the system to operate over the temperature range specified in 3.8.1.

* 4.3.13 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract. The manufacturer shall maintain the vehicle in an as new condition for the duration of the contract. Release of the first vehicle for shipment prior to completion of the contract shall be at the discretion of the local Government Quality Assurance Representative (QAR).

* 4.3.14 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

* 4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting the performance requirements contained herein. All semitrailers under contract shall be tested as specified in 4.3.2. For each vehicle under contract, the contractor shall

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make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The semitrailers covered by this specification are intended or general non-tactical use by the Government in the bulk loading, transportation and dispensing of nonpotable water. Semitrailers will operate over both improved and unimproved roads.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size and class of vehicle required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate military service for painting and marking (see 3.1.1.1 and 3.1.1.2).
(e) Lifting attachments and tiedown devices, if required (see 3.1.1.7).
(f) Pintle assembly, if required (see 3.1.1.9).
(g) Independent 24-volt dc lighting system, if required (see 3.1.1.10.1).
(h) Interconnected 24-volt dc lighting system, if required (see 3.1.1.10.2).
(i) Air suspension system, if required (see 3.8.1).
(j) Spare wheel or rim, if required (see 3.9.3).
(k) Spare tire assembly, if required (see 3.9.4).
(1) Tools, if required (see 3.9.5).
(m) Air/hydraulic powered landing gear, if required (see 3.12.2).

6.3 Subject term (key word) listing.
Air suspension system
Bulk transportation of nonpotable water Diesel engine
DOT 406
General purpose water tank
Hose
MC 306
Non-tactical vehicle (NTV)
Semitrailer tank.

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6.4 Changes from previous issue. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

Custodians:
Army - AT

Preparing activity:
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